

SEBASTIAN BARTLING

(+49) 170 108 109 0
sebastian-bartling@hotmail.de
<https://sebastianpbartling.github.io/sb.github.io/>

POSITIONS	Department of Mathematics <i>Wissenschaftlicher Mitarbeiter</i> • Mentor: Prof. Vytautas Paskunas • Research area: Arithmetic Geometry	Essen, Germany 2022 - 2028
EDUCATION	Department of Mathematics, Sorbonne University <i>Ph.D. in Mathematics</i> • Advisor: Prof. Laurent Fargues • Research area: Arithmetic Geometry Department of Mathematics, Paris Sud <i>M2 in Arithmetic, Analysis and Geometry</i> • Advisor: Laurent Fargues Department of Mathematics, Essen <i>M1 in Mathematics</i> • Specialization in number theory and algebraic geometry Department of Mathematics, Bielefeld <i>Bachelor in Mathematics and Philosophy</i>	Paris, France 2018 - 2022 Orsay, France 2017 - 2018 Essen, Germany 2016 - 2017 Bielefeld, Germany 2013 - 2016
PUBLICATIONS AND PREPRINTS	1. Sebastian Bartling. The universal special formal $\mathcal{O}_D$-module for $d = 2$, 2022 (accepted under revision at MRL). 2. Sebastian Bartling. $\mathcal{G}$-μ-displays and local Shtuka, 2022. 3. Sebastian Bartling. Sur la cohomologie étale de la courbe de Fargues-Fontaine, 2022. 4. Sebastian Bartling, Manuel Hoff. Moduli spaces of nilpotent displays, 2024 (IMRN) 5. Sebastian Bartling, Kazuhiro Ito. Vanishing of Brauer groups of moduli stacks of stable curves, 2024, Forum of Mathematics, Sigma, 6. Sebastian Bartling, Kazuhiro Ito. Close fields, affine Springer fibers and fundamental lemmas, 2026, revision requested at Forum of Mathematics, Pi (since 11.9.2026), 7. Sebastian Bartling, Andreas Mihatsch, Families of intersection numbers over close fields, 2026 (available on my website), 8. Sebastian Bartling, Rıżacan Çilog�lu, Motivic parahoric Hecke categories in mixed and equal characteristics, 2026, submitted to Journal de l’Institut de math�matique de Jussieu	
PROJECTS IN PROGRESS	Integral models of local Shimura-varieties via sheared displays <i>joint work in progress with Manuel Hoff</i> The arithmetic Gan–Gross–Prasad conjecture over function fields for single-legged hermitian shtukas <i>joint work in progress with Kazuhiro Ito and Andreas Mihatsch</i>	2026 2026

SELECTED RESEARCH TALKS	Séminaire GRFA (Jussieu, Summer 2022)
	Oberwolfach Research Institute (February 2023)
	Seminar on Ramification theory (Beijing university, Spring 2023)
	Number theory seminar (Morningside center for Mathematics, Spring 2023)
	Arithmetic and Algebraic Geometry seminar (UVA, Amsterdam, Winter 2024)
	Number theory seminar (Tohoku University, Spring 2025)
	Representation theory and number theory seminar (IAS, Zhejiang University, Spring 2025)
	Algebra and number theory seminar (Wuppertal, Winter 2025)
RESEARCH VISITS	Morningside center for Mathematics, 5 weeks, Winter 2019
	Beijing university, 2 weeks, Spring 2023
	Tohoku university, 2 weeks, Spring 2025
	IAS, Zhejiang university, 1 week, Spring 2025
	Zhejiang university, 2 weeks, Spring 2026
	Tohoku university, 2 weeks, Spring 2026
SKILLS	Languages: German, English, French.
	Programming: Python, C++, MATLAB.
ACADEMIC SERVICES	Referee for: <i>Journal für die reine und angewandte Mathematik, Forum of Mathematics, Pi</i>
	Organization of: <i>ENTR Workshop in Essen, 2024</i>